Peatlands in Flux: Flora Response to Snow Depth in Northern Scandinavia

Laurie Quincey[1], Fabien Maussion[1]

1 University of Bristol, Beacon House, Queens Road, Bristol, BS8 1QU, UK

University of Bristol

Introduction

Understanding Arctic flora response to climate is critical for ecology, society, and climate research, yet the influence of snow depth fluctuations on Arctic flora productivity remains poorly understood on climate timescales. With less ubiquitous Arctic greening trends[1, 2] and weakened temperature-productivity relationships[3, 4], snow may be an important influence over productivity[5-7].

However, under-validation and a reliance on satellite data decontextualises research from site-specific ecology[8]. Data fusion approaches may remedy this, retaining site-specific context over climate timescales.

This study reconstructed 25 years (2000-2025) of sub-weekly gross primary productivity (**GPPe**) at Abisko-Stordalen, Sweden, using random forest regressors to model site-specific flux tower-derived GPP (**GPPf**) from satellite data.

Research Questions

- Can random forest regression convert optical satellite data to flux tower GGPf?
- How does snow depth affect gross primary productivity (GPP)?
- How does the study method compare to traditional satellite measures of GPP?

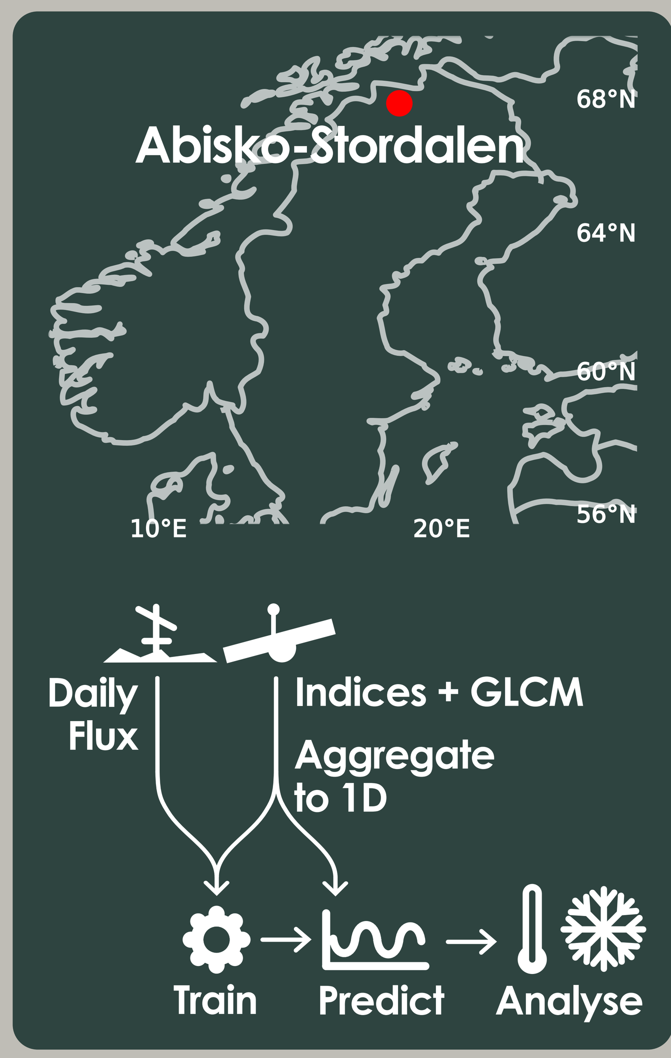
Methodology

Processing

- Calculate indices and texture on MODIS[9, 10], Landsat 4-9[11-15], and Sentinel-2[16, 17].
- Aggregate pixels within 200 m of flux towers.
- Calculate daily GPPf from ICOS Abisko-Stordalen Palsa and Grassland flux towers[18, 19].
- Train site-specific random forest models on satellite and GPPf relationship.
- Predict sub-weekly GPPe and calculate annual & seasonal GPPe.
- Prepare warmth index (WI) and snow depth from SMHI observations[20].

Analysis

- Z-score GPPe & NDVI and find inter-quartile range.
- Theil-sen slope/Mann-Kendall test on annual GPPe.
- Ordinary least squares model on seasonal GPPe & WI and GPPe WI residuals & snow depth.



Results

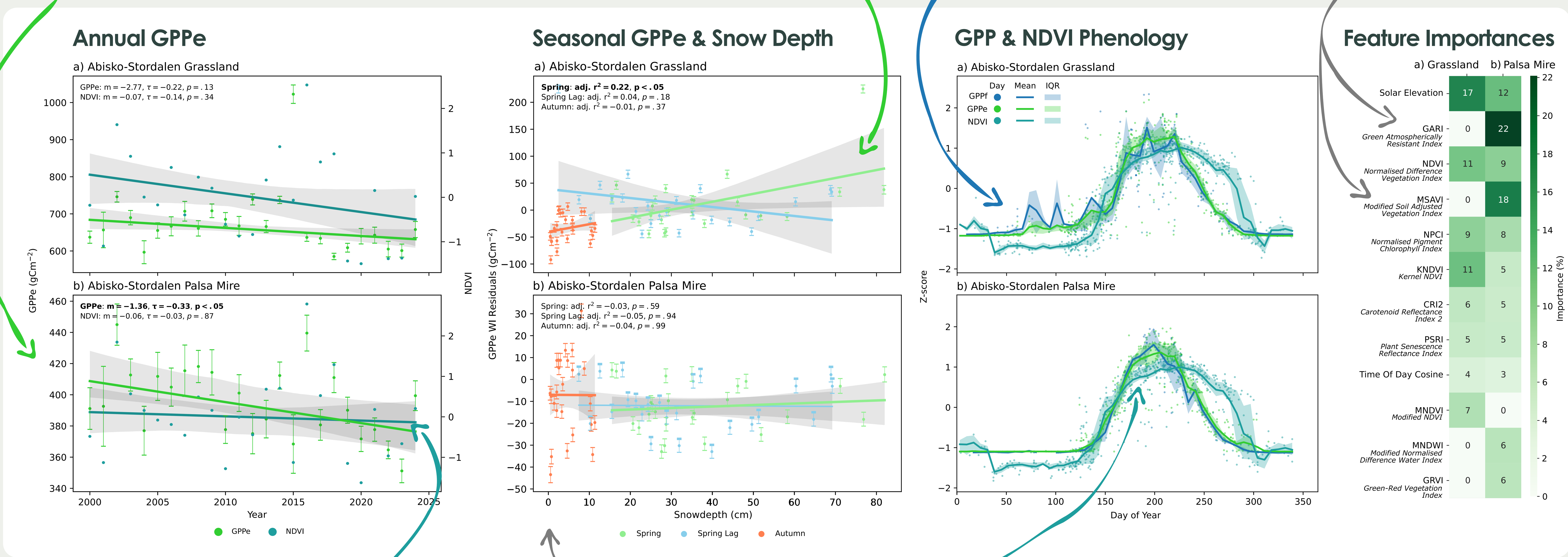
GPP = Gross Primary Production; GPPe = GPP Model Estimate; GPPf = Flux Tower GPP

Abisko-Sto. Pasla Mire reported a significant decrease in annual **GPPe** of **~34 gCm^{-2} over 25 years** (~8-9%)

Spring snow depth had a significant positive relationship to **GPPe** at Abisko-Sto. Grassland...

Study **GPPe** recreated **GPPf** phenology faithfully but did not capture small-scale variation

GARI & **MSAVI** indices were unique to the Palsa Mire model



Feature	a) Grassland	b) Palsa Mire
Solar Elevation	17	12
GARI	0	22
NDVI	11	9
MSAVI	0	18
NPCI	9	8
KNDVI	11	5
CRI2	6	5
PSRI	5	5
Time Of Day Cosine	4	3
MNDVI	7	0
MNDWI	0	6
GRVI	0	6

In contrast, **NDVI** reported no significant trend at Abisko-Sto. Palsa Mire

...**but** no snow depth trend at the Palsa Mire

NDVI phenology captures spring growth and **but** undershoots peak **GPPf** and overshoots autumn **GPPf**

Model Validation

a) Grassland: R^2=0.72, MAE=4.04 gCm^{-2}

b) Palsa Mire: R^2=0.80, MAE=1.63 gCm^{-2}

Discussion

At Abisko-Sto. Grassland, GPPe increased with snow depth. This was attributed to increased soil insulation causing nutrient release and extended ablation seasons. This may promote flora that benefit from snowmelt and nutrient rich conditions[21-25]. **Conversely, the Palsa Mire reported no snow depth relationship.** As raised palsas are susceptible to wind removal of snow, flora may remain temperature-limited in spring from reduced snow insulation[26].

The Palsa Mire reported a long-term GPP decrease. This contrasted with warming trends[27] and increased productivity reported by NDVI-based analysis at higher latitudes[6]. This discrepancy may occur as permafrost under the Palsa Mire is melting, causing collapse and inundation[28, 29]. The resultant anoxic conditions may decrease GPP[23-25] and carbon sequestration as reported at the Palsa Mire and nearby Abisko-Storflaket Mire[30, 31].

Validation showed the study method was suitable as GPPe models faithfully reproduced GPPf phenology. Performance was better than NDVI which is a commonly used Arctic Greening metric[32, 33]. As a result, the discrepancy between long-term GPPe and NDVI trends at the Palsa Mire suggests **NDVI may perform poorly on mosaicked landscapes and peatland flora**, echoing other studies[34-36]. As indices such as GARI were important model features, wavelengths not used by NDVI[37, 38] may contain information important for estimating GPP. Therefore, this study suggests **multi-index approaches may better characterise GPP across diverse Arctic flora** compared to relying solely on NDVI.

Conclusion & Implications

- Random forest models can convert satellite data to flux tower GPP.
- Snow depth raises GPP, perhaps by soil insulation & snow melt.
- Thermokarst development may reduce Arctic carbon sequestration.
- NDVI skews phenology and can distort long-term GPP trends.
- NDVI may be unreliable for Arctic Greening and peatland monitoring.
- This study recommends multi-index approaches.

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Author Contributions: L.Q. conceived of and carried out this study, F.M. supervised and advised.

lrquincey@gmail.com github.com/lauriequincey linkedin.com/in/lauriequincey